THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. One angle of a linear pair is 4 times the other. The larger angle is: 1

(a) 164° **(b)** 154° **(c)** 134° **(d)** 144°

2. An angle is equal to its complement. The angle is: 1

(a) 45° **(b)** 60° **(c)** 90° **(d)** 30°

3. Two supplementary angles are in the ratio $1 : 5$. The larger angle is: 1

(a) 170° **(b)** 140° **(c)** 150° **(d)** 160°

4. The supplement of the complement of 65° is: 1

(a) 165° **(b)** 155° **(c)** 175° **(d)** 145°

5. The reflex angle that goes with an angle of 25° (so that together they make a complete turn) is: 1

(a) 325° **(b)** 355° **(c)** 335° **(d)** 345°

6. Two complementary angles differ by 40°. The larger angle is: 1

(a) 75° **(b)** 55° **(c)** 65° **(d)** 85°

7. Two angles of a linear pair differ by 100°. The smaller angle is: 1

(a) 60° **(b)** 30° **(c)** 40° **(d)** 50°

8. One angle of a linear pair is 5 times the other. The larger angle is: 1

(a) 150° **(b)** 160° **(c)** 170° **(d)** 140°

9. **Assertion (A):** A transversal cuts two lines making alternate interior angles of 65° each. The lines are parallel. 1

Reason (R): If a pair of alternate interior angles is equal, the two lines are parallel.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The angles of a linear pair are 110° and 70°. 1

Reason (R): The sum of all angles around a point is 360°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two supplementary angles differ by 46°. Find both angles and the complement of the smaller one. **2**

12. Find the angle whose supplement is 7 times its complement. **2**

13. An angle is 18° more than 3 times its complement. Find the angle. **2**

14. Two supplementary angles differ by 72°. Find both angles and the complement of the smaller one. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Angles on one side of a straight line at a point are 30°, 30°, 70° and a fourth angle. Find the fourth angle and name the kind of each angle. **3**

16. Two supplementary angles are in the ratio $3 : 7$. Find the complement of the smaller angle and the supplement of the larger angle. **3**

17. Two adjacent angles form a linear pair and one of them is 130°. Find the other angle, the half of each angle and the angle between the two bisectors. **3**

OR

Angles on one side of a straight line at a point are 55°, 25°, 65° and a fourth angle. Find the fourth angle and name the kind of each angle. **3**

18. Two supplementary angles are in the ratio $1 : 14$. Find the complement of the smaller angle and the supplement of the larger angle. **3**

OR

Two adjacent angles form a linear pair and one of them is 150°. Find the other angle, the half of each angle and the angle between the two bisectors. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A ladder leans against a vertical wall. It makes an angle of 70° with the horizontal ground. The wall makes 90° with the ground. **5**

(i) What is the angle between the ladder and the wall?

(ii) What is the supplement of 70°?

(iii) Are 40° and 50° complementary?

(iv) An angle is 20° more than its complement. Find the angle and its complement.

20. Three angles on one side of a straight line at a point are in the ratio $3 : 4 : 5$. Find the three angles and name the kind of each. **5**

OR

A transversal cuts two parallel lines and two interior angles on the same side are in the ratio $1 : 17$. Find these two angles and say how many of the eight angles measure 10° and how many 170°. **5**

END OF QUESTION PAPER — Total: 40 Marks